

COAL Lab (EL2003)

Computer organization and assembly language Lab

Date: December 20, 2025

Lab Instructor(s)

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Final Exam

Total Time (Hrs): 2.5

Total Marks: 45

Total Questions: 2

Roll No

Section

Student Signature

Instructions: It is an open book exam. Only textbook "Assembly Language Programming Lecture Notes by Bilal Hashmi" is allowed. ANY NOTES OR ATTACHMENTS ARE NOT ALLOWED.

Submission Folder Path: \\exam\Fall 2025 Lab Final\COAL\your section\Submission

Assembly Programming Tools Folder: \\exam\Fall 2025 Lab Final\COAL\Software\Assembly Programming Tools

Copy / paste Assembly Programming Tools to your Desktop in order to run/use it.

Important Instructions (Please read them before attempting the exam):

- Put your all .asm files in single folder name it **Roll_no_Section_Final(25L_1234)** and check the size of file and folder. It should not be 0 Kb.
- You are not allowed to use Internet or any Gadgets including USB device, cell phones.
- Plagiarism will result in **F grade** in lab
- Protect your code; if copied, you're responsible.
- No questions allowed; make assumptions and proceed.
- Read questions carefully.

Name of your file should be "YourRollNo_Question#.asm". Try to submit running code (without any errors). **YOUR FINAL WILL BE MARKED ZERO IF YOUR_ROLL_NO.ASM IS MISSING IN SUBMISSION FOLDER**

CLO # 3: Apply the knowledge of Intel x86 architecture to develop moderately complex and well-modularized assembly programs.

Question #1 [30 Marks]:

Write an 8088 assembly language program that simulates a traffic light system displayed on the screen as a colored square. The square must represent the traffic light using **Green, Yellow, and Red** colors, and it must remain visible on the screen at all times.

(a)

When the program starts, the traffic light should operate in **automatic mode**. In this mode, the **color of the square must change every 5 seconds** in a continuous **Red → Yellow → Green** sequence. The timing for color changes must be controlled using a **timer interrupt**, demonstrating concurrent execution between display and timing logic. *In the pictures below, has all colors red, yellow, and green in sequence. Due to printing, they might be visible in black and white. Your program should display in colors.*



(b) While the traffic light is running in automatic mode, pressing the ENTER key should immediately switch the program into a password input mode handled through the keyboard hardware interrupt. The user must enter a fixed four-digit numeric password (1122), and only numeric keys (0–9) are allowed. After exactly four digits are entered, the input must be compared with the stored password.

If the password is incorrect, an “Access Denied” message should be displayed and the program must return to automatic traffic light operation. If the password is correct, an “Access Granted” message should be displayed and the program should switch to manual control mode, at which point the automatic timer-based light switching must stop.

(c) In manual control mode, the color of the square must change every 3 seconds in a continuous Red → Yellow → Green sequence. Pressing the ESC key should exit manual mode and return the program to automatic mode, and pressing the ESC key again while in automatic mode should terminate the program.

Question # 2 [15 Marks]::

Write a program that displays the string “HELLO WORLD” moving horizontally on the screen. The string should initially appear at row 0 and column 12 and move one position to the right on each update. When the string reaches column 60, it should reverse direction and move toward the left. Upon reaching column 12, the direction should again be reversed, creating a continuous back-and-forth motion.

The program should behave in a multitasking manner such that each key press starts a new moving instance of the string on a incremented row but same column number, with a maximum of 12 simultaneous moving strings allowed on the screen at any time. Each moving string must operate independently, bouncing between the left and right limits without interfering with others.

While the strings are moving, their previous positions must be cleared by writing blank spaces so that no trails remain on the screen. The program should continue running until the ESC key is pressed, after which all movement should stop and the program should terminate gracefully.

ASCII & Scan Codes

Key	0	1	2	3	4	5	6	7	8	9	ENTER	ESC	G	Y	R
ASCII (Hex)	30h	31h	32h	33h	34h	35h	36h	37h	38h	39h	0Dh	1Bh	47h	59h	52h
Scan Code (Hex)	0Bh	02h	03h	04h	05h	06h	07h	08h	09h	0Ah	1Ch	01h	22h	15h	13h

Important Instructions:

- Properly comment your code and write functions where needed. Freely use global variables, if required.